

Module 2

Management Accounting

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Budgeted statement of profit or loss for the next two years using absorption costing system

	<u>Year 1</u>	<u>Year 2</u>
	<u>HK\$'000</u>	<u>HK\$'000</u>
Sales	<u>72,000</u>	<u>96,000</u>
Less: Production costs of sales		
Opening inventory	-	20,300
Add: Variable production costs (W1)	65,000	52,000
Fixed production overhead costs	<u>7,500</u>	<u>6,000</u>
	72,500	78,300
Less: Closing inventory	<u>20,300</u>	<u>8,700</u>
	52,200	69,600
Under/ (Over) absorbed fixed production overhead costs (W2)	<u>(1,200)</u>	<u>300</u>
	<u>51,000</u>	<u>69,900</u>
Gross profit	<u>21,000</u>	<u>26,100</u>
Less: Variable non-production overhead cost (W3)	3,600	4,800
Fixed non-production overhead costs	<u>5,000</u>	<u>5,000</u>
	<u>8,600</u>	<u>9,800</u>
Net profit	<u>12,400</u>	<u>16,300</u>

Answer 1(b)

Budgeted statement of profit or loss for the next two years using marginal costing system

	<u>Year 1</u>	<u>Year 2</u>
	<u>HK\$'000</u>	<u>HK\$'000</u>
Sales	<u>72,000</u>	<u>96,000</u>
Less: Variable production costs of sales		
Opening inventory	-	18,200
Add: Variable production costs (W1)	<u>65,000</u>	<u>52,000</u>
	65,000	70,200
Less: Closing inventory	<u>18,200</u>	<u>7,800</u>
	46,800	62,400
Variable non-production overhead costs (W3)	<u>3,600</u>	<u>4,800</u>
	<u>50,400</u>	<u>67,200</u>
Contribution margin	<u>21,600</u>	<u>28,800</u>
Less: Fixed production overhead costs (W4)	6,300	6,300
Fixed non-production overhead costs	<u>5,000</u>	<u>5,000</u>
	<u>11,300</u>	<u>11,300</u>
Net profit	<u>10,300</u>	<u>17,500</u>

Workings:

- (W1) Year 1: $\text{HK}\$(80 + 20 + 30) \times 500,000 = \text{HK}\$65,000,000$
Year 2: $\text{HK}\$(80 + 20 + 30) \times 400,000 = \text{HK}\$52,000,000$
- (W2) Year 1: $(\text{HK}\$15 \times 420,000) - (\text{HK}\$15 \times 500,000) = (\text{HK}\$1,200,000)$
Year 2: $(\text{HK}\$15 \times 420,000) - (\text{HK}\$15 \times 400,000) = \text{HK}\$300,000$
- (W3) Year 1: $\text{HK}\$10 \times 360,000 = \text{HK}\$3,600,000$
Year 2: $\text{HK}\$10 \times 480,000 = \text{HK}\$4,800,000$
- (W4) Year 1 & 2: $\text{HK}\$15 \times 420,000 = \text{HK}\$6,300,000$

Answer 1(c)

Reconciliation of the budgeted profits:

	<u>Year 1</u>	<u>Year 2</u>
	<u>HK\$'000</u>	<u>HK\$'000</u>
Absorption costing profit	12,400	16,300
Year 1 - Adjustment for fixed overheads in increase in closing inventory (140,000 units $\times$ HK\$15)	(2,100)	-
Year 2 - Adjustment for fixed overheads in decrease in closing inventory (80,000 units $\times$ HK\$15)	-	1,200
Marginal costing profit	<u>10,300</u>	<u>17,500</u>

Answer 1(d)

The reasons for using a marginal costing system instead of an absorption costing system for internal decision making include:

- Marginal costing is simple to operate as it does not require complex apportionments and calculation of overhead absorption rates.
- Marginal costing ties in with standard costing and flexible budgeting more effectively.
- When using marginal costing, net profit is affected by changes in sales as opposed to production. This provides a more realistic assessment of success or failure during a particular period.
- Net profit calculated under marginal costing is not affected by changes in production volume. Marginal costing makes it easier to understand the impact of fixed and variable costs on net profit.
- Marginal costing is consistent with the cost-volume-profit relationship and with the incremental analysis of costs.
- Marginal costing produces statements of profit or loss that are closer to net cash flows than absorption costing.

(Any three reasons. Other reasonable answers are also accepted.)

Answer 2(a)

Shortage of labour hours:

	Aluminium	Plywood	Total
Labour hours per unit	3 hrs	4 hrs	
Sales demand in units	12,000	10,000	
Labour hours needed	<u>36,000 hrs</u>	<u>40,000 hrs</u>	76,000 hrs
Labour hours available			<u>62,000 hrs</u> <u>14,000 hrs</u>

Answer 2(b)

Contribution margin per unit of limiting factor (labour hour):

	Aluminium HK\$	Plywood HK\$
Unit selling price	<u>2,900</u>	<u>4,000</u>
Unit variable costs:		
Direct materials	850	1,400
Direct labour	300	400
Variable production overhead	<u>250</u>	<u>600</u>
	<u>1,400</u>	<u>2,400</u>
Unit contribution margin	<u>1,500</u>	<u>1,600</u>
Labour hours per unit	3 hrs	4 hrs
Contribution margin per labour hour	HK\$500	HK\$400
Production priority	1 st	2 nd

	Aluminium	Plywood	Total
Optimal production plan in units (W1)	12,000	6,500	
Labour hours required (W2)	36,000	26,000	62,000
Unit contribution margin	HK\$1,500	HK\$1,600	
	HK\$'000	HK\$'000	HK\$'000
Total contribution margin	<u>18,000</u>	<u>10,400</u>	28,400
Less: Fixed production overhead (W3)			<u>12,200</u>
Profit			<u>16,200</u>

Workings:

(W1) Production for Plywood: $[(62,000 - 36,000) \div 4]$ units = 6,500 units(W2) Labour hours required for Plywood: $(62,000 - 36,000)$ hours = 26,000 hours(W3) $\text{HK\$}600 \times 12,000 + \text{HK\$}500 \times 10,000 = \text{HK\$}12,200,000$

Answer 2(c)

Ways for FML to overcome labour shortage:

- Outsource the production to other producers.
- Consider/ enhance automation of production to reduce the requirement for labour.
- Redesign the production process to reduce the requirement for labour.
- Raise labour efficiency through incentive and training schemes.
- Import labour with the same skills from other countries.

(Any three ways. Other reasonable answers are also accepted.)

Answer 3(a)

Labour costs are semi-variable costs (or mixed costs, or step costs).

Semi-variable costs are costs that include both fixed and variable components.

Answer 3(b)

The budgeted production costs for output of 25,000 units:

	HK\$
Material costs (W1)	750,000
Labour costs (W2)	1,450,000
Production overhead costs	<u>1,500,000</u>
Production costs	<u>3,700,000</u>

Workings:

(W1) $(\text{HK\$}600,000 \div 20,000) \times 25,000 = \text{HK\$}750,000$

(W2) Unit variable cost: $(\text{HK\$}1,400,000 - \text{HK\$}1,200,000) \div (24,000 - 20,000) = \text{HK\$}50$

Fixed cost: $\text{HK\$}1,200,000 - \text{HK\$}50 \times 20,000 = \text{HK\$}200,000$

Total labour costs for 25,000 units: $\text{HK\$}200,000 + \text{HK\$}50 \times 25,000 = \text{HK\$}1,450,000$

Answer 3(c)

Production budget in units:

	January	February	March
Sales	30,000	28,000	40,000
Closing inventory (W3)	<u>11,200</u>	<u>16,000</u>	<u>14,000</u>
	41,200	44,000	54,000
Opening inventory (W4)	<u>(12,000)</u>	<u>(11,200)</u>	<u>(16,000)</u>
Good output required	29,200	32,800	38,000
Normal loss (W5)	<u>3,244</u>	<u>3,644</u>	<u>4,222</u>
	<u><u>32,444</u></u>	<u><u>36,444</u></u>	<u><u>42,222</u></u>

Workings:

- (W3) January: $28,000 \times 40\% = 11,200$
February: $40,000 \times 40\% = 16,000$
March: $35,000 \times 40\% = 14,000$
- (W4) January: $30,000 \times 40\% = 12,000$
February: $28,000 \times 40\% = 11,200$
March: $40,000 \times 40\% = 16,000$
- (W5) January: $29,200 \times (10\% \div 90\%) = 3,244$
February: $32,800 \times (10\% \div 90\%) = 3,644$
March: $38,000 \times (10\% \div 90\%) = 4,222$

Answer 4(a)

Residual income (RI): $\text{HK\$}5,760,000 - \text{HK\$}32,000,000 \times 15\% = \text{HK\$}960,000$

The positive RI indicates that the performance is acceptable and the return is higher than the required rate of return.

Answer 4(b)

The advantages include:

- While financial metrics focus on an accounting period, non-financial measures can be used to track performance over a much longer period of time and without necessarily comparing how that metric performed in relation to competitors, thus providing a window into the progress of accomplishing long-term company strategies.
- Non-financial measures may allow companies to track intangible assets such as intellectual capital or customer loyalty. Companies have grown to value performance measurement indicators that track innovation, capabilities of management, employee relations, brand quality, etc.
- Non-financial measures may be better indicators of future performance than financial measures. For example, investment in research is more likely to provide a better indication of how companies will perform in the future than sales and profit in the last quarter.
- Non-financial measures may provide measures that are not tied to the many factors that can distort financial measures, such as changes in economic circumstances, which may have little or nothing to do with company performance.

(Any three advantages. Other reasonable answers are also accepted.)

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